| | | | | |
|---|---|---|---|---|
| **8** | **(a)** | energy is given out / released on formation of the $\alpha$-particle (or reverse argument) | M1 | |
| | | either $E = mc^2$ so mass is less | | |
| | | or reference to mass-energy equivalence | A1 | [2] |
| | **(b) (i)** | mass change = 18.00567 u – 18.00641 u | C1 | |
| | | $= 7.4 \times 10^{-4}$ u *(sign not required)* | A1 | [2] |
| | **(ii)** | energy = $c^2\Delta m$ | | |
| | | $= (3.0 \times 10^8)^2 \times 7.4 \times 10^{-4} \times 1.66 \times 10^{-27}$ | C1 | |
| | | $= 1.1 \times 10^{-13}$ J | A1 | [2] |
| | | (*allow use of* u = $1.67 \times 10^{-27}$ kg) | | |
| | | (*allow method based on* 1u *equivalent to* 930 MeV *to* 933 MeV) | | |
| | **(iii)** | *either* mass of products greater than mass of reactants | M1 | |
| | | this mass/energy provided as kinetic energy of the helium-4 nucleus | A1 | |
| | | *or* both nuclei positively charged | (M1) | |
| | | energy required to overcome electrostatic repulsion | (A1) | [2] |

## Section B